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Inventor(s)	Song; Minghui et al.

Double-microphone array echo eliminating method, device and electronic equipment

Abstract

Embodiments of the present application relate to a dual-microphone array echo cancellation method, device and electronic equipment, comprises: the acquired distal signal, the first proximal signal and the second proximal signal are processed by linear filtering; obtain initial suppression gain factors and variable step-size factors of the first near-end signal and the second near-end signal; performing residual echo suppression on the first error frequency spectrum and the second error frequency spectrum by using an adaptive zero-pole echo canceller to obtain a target frequency spectrum signal; performing sub-band range selection on the initial suppression gain factors to obtain a smooth factor, and performing full-band smoothing and exponential operation to obtain a secondary suppression gain factor; performing filtering processing on the target frequency spectrum signal by using the secondary suppression gain factor to obtain a target near-end voice signal.

Inventors:	Song; Minghui (Shenzhen, CN), Lin; Jinhong (Shenzhen, CN)
Applicant:	Shenzhen Bluetrum Technology Co., Ltd. (Shenzhen, CN)
Family ID:	82509355
Assignee:	Shenzhen Bluetrum Technology Co., Ltd. (Shenzhen, CN)
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Primary Examiner: Phan; Hai

Assistant Examiner: Iskender; Alvin

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) The present application claims the benefit of Chinese Patent Application No. 202210470223.9 filed on Apr. 28, 2022, the contents of which are incorporated herein by reference in their entirety.

TECHNICAL FIELD

(2) Embodiments of the present application relate to the technical field of echo cancellation, and in particular, relate to a dual-microphone array echo cancellation method, a dual-microphone array echo cancellation device and electronic equipment.

BACKGROUND

(3) Echo cancellation (AEC) technology is widely used in modern communication electronic equipments. In the process of a phone conversation between two parties, the near-end microphone will collect the sound of the speaker so that the far end can hear its own echo. Therefore, the performance of echo cancellation seriously affects the quality of phone conversation between the two parties.

SUMMARY

(4) An objective of embodiments of the present application is to provide a dual-microphone array echo cancellation method, device and electronic equipment, which can effectively improve the echo cancellation effect and improve the quality of voice call.

(5) An embodiment of the present invention provides a dual-microphone array echo cancellation method, and the method comprises: acquiring far-end signals generated by an electronic equipment during a phone conversation, and a first near-end signal and a second near-end signal of a dual microphone array; performing linear filtering processing on the far-end signals, the first near-end signal and the second near-end signal to obtain a first initial error frequency spectrum and a second initial error frequency spectrum; performing centroid fitting on the first initial error frequency spectrum and the second initial error frequency spectrum respectively to obtain initial suppression gain factors of the first near-end signal and the second near-end signal, and using the initial suppression gain factors as variable step-size factors for linear filtering processing of the next frame; performing residual echo suppression on the first error frequency spectrum and the second error frequency spectrum by using an adaptive zero-pole echo canceller to obtain a target frequency spectrum signal; after performing sub-band range selection on the initial suppression gain factors, obtaining a smooth factor, and performing full-band smoothing and exponential operation to obtain a secondary suppression gain factor; performing filtering processing on the target frequency spectrum signal by using the secondary suppression gain factor to obtain a target near-end voice signal.

(6) An embodiment of the present invention provides an echo cancellation device for a dual microphone array, and the device comprises: a signal acquisition module, being configured to acquire far-end signals generated by an electronic equipment during a phone conversation, and a first near-end signal and a second near-end signal of a dual microphone array; a linear filtering module, being configured to perform linear filtering processing on the far-end signals, the first

near-end signal and the second near-end signal to obtain a first initial error frequency spectrum and a second initial error frequency spectrum; a centroid fitting module, being configured to perform centroid fitting on the first initial error frequency spectrum and the second initial error frequency spectrum respectively to obtain initial suppression gain factors of the first near-end signal and the second near-end signal, and use the initial suppression gain factors as variable step-size factors for linear filtering processing of the next frame; a residual echo suppression module, being configured to perform residual echo suppression on the first error frequency spectrum and the second error frequency spectrum by using an adaptive zero-pole echo canceller to obtain a target frequency spectrum signal; a smoothing module, being configured to obtain a smooth factor and perform full-band smoothing and exponential operation to obtain a secondary suppression gain factor after performing sub-band range selection on the initial suppression gain factor; a target acquisition module, being configured to perform filtering processing on the target frequency spectrum signal by using the secondary suppression gain factor to obtain a target near-end voice signal.

(7) An embodiment of the present invention provides an electronic equipment, which comprises: a dual microphone array and at least one processor, and a memory, wherein the memory is communicatively connected with the at least one processor, and the memory stores instructions executable by the at least one processor, and the instructions, when executed by the at least one processor, cause the electronic equipment to execute the method of any of the aspects described above.

(8) An embodiment of the present invention provides a nonvolatile computer readable storage medium, wherein the computer readable storage medium stores computer-executable instructions which, when executed by an electronic equipment, cause the electronic equipment to execute the method of any of the aspects described above.

(9) According to the dual-microphone array echo cancellation method, device and electronic equipment provided by the embodiments of the present invention, far-end signals generated by an electronic equipment during a phone conversation as well as a first near-end signal and a second near-end signal of a dual microphone array are acquired, then preliminary filtering processing is performed on the far-end signals and the near-end signals of the dual microphone array, e.g., linear filtering processing may be performed on the far-end signals, the first near-end signal and the second near-end signal to obtain a first initial error frequency spectrum and a second initial error frequency spectrum so as to preliminarily cancel the echo; next, centroid fitting is performed on the first initial error frequency spectrum and the second initial error frequency spectrum respectively to obtain initial suppression gain factors of the first near-end signal and the second near-end signal, and the initial suppression gain factors are used as variable step-size factors for linear filtering processing of the next frame; residual echo suppression is performed on the first error frequency spectrum and the second error frequency spectrum by using an adaptive zero-pole echo canceller to obtain a target frequency spectrum signal so as to achieve residual echo suppression; after performing sub-band range selection on the initial suppression gain factor, a smooth factor is obtained, and full-band smoothing and exponential operation are performed to obtain a secondary suppression gain factor; filtering processing is performed on the target frequency spectrum signal by using the secondary suppression gain factor to achieve secondary suppression gain filtering and obtain a target near-end voice signal, thereby effectively improving the echo cancellation effect and improving the quality of voice call.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) One or more embodiments are illustrated by pictures in corresponding attached drawings, and this does not constitute limitation on the embodiments. Elements with the same reference numerals

in the attached drawings are shown as similar elements, and the pictures in the attached drawings do not constitute scale limitation unless otherwise stated particularly.

(2) FIG. 1 is a schematic block diagram of an electronic equipment according to an embodiment of the present invention.

(3) FIG. 2 is a circuit structural diagram of an electronic equipment according to an embodiment of the present invention.

(4) FIG. 3 is a schematic structural diagram of a dual microphone array of an electronic equipment according to an embodiment of the present invention.

(5) FIG. 4 is a schematic flowchart diagram of one embodiment of a dual-microphone array echo cancellation method according to an embodiment of the present invention.

(6) FIG. 5 is a schematic structural diagram of one embodiment of a dual-microphone array echo cancellation device according to the present invention.

(7) FIG. 6 is a schematic view of the hardware structure of a controller in one embodiment of the electronic equipment according to the present invention.

DETAILED DESCRIPTION

(8) The present application will be described in detail hereinafter with reference to specific embodiments. The following embodiments will facilitate the further understanding of the present application by those skilled in the art, but are not intended to limit the present application in any way. It shall be noted that, those of ordinary skill in the art can make several modifications and improvements without departing from the concept of the present application. All these modifications and improvements belong to the scope claimed in the present application.

(9) In order to make objectives, technical solutions and advantages of the present application clearer, the present application will be further described in detail hereinafter with reference to attached drawings and embodiments. It shall be appreciated that, the specific embodiments described herein are only used to explain the present application, and are not used to limit the present application.

(10) It shall be noted that, all features in the embodiments of the present application may be combined with each other without conflict, and all the combinations are within the scope claimed in the present application. In addition, although functional module division is made in the schematic diagrams of the device and logical sequences are shown in the flowchart diagrams, in some cases, the steps shown or described can be executed with module division and sequences different from those in the schematic diagrams of the device and the flowchart diagrams. Furthermore, words such as “first”, “second”, and “third” used herein do not limit the data and execution order, but only distinguish same or similar items with basically the same functions and effects.

(11) Unless otherwise defined, all technical and scientific terms used in this specification have the same meanings as commonly understood by those skilled in the art of the present application. In this specification, the terms used in the specification of the present application are only for the purpose of describing specific embodiments, and are not intended to limit the present application. The term “and/or” used in this specification comprises any and all combinations of one or more associated items listed.

(12) In addition, the technical features involved in various embodiments of the present application described below can be combined with each other as long as they do not conflict with each other.

(13) The dual-microphone array echo cancellation method and device provided by the embodiments of the present application may be applied to electronic equipments, and the electronic equipments may be smart phones, Bluetooth headphones, wired headphones and other equipments used for phone communication. Taking the Bluetooth headphones as an example, the Bluetooth headphones are connected via Bluetooth with the smart phone which is required to make a conversation, and a user can make the conversation by wearing the Bluetooth headphones.

(14) As shown in FIG. 1 and FIG. 2, an electronic equipment **100** comprises a controller **11**, a

speaker **12** and a dual microphone array **13**, and the controller **11** is connected with the speaker **12** and the dual microphone array **13**. The speaker **12** is used for playing sounds which are far-end signals. The dual microphone array **13** is used for collecting sounds, and the sounds collected by the dual microphone array **13** comprise the voice of the user, echo and background noise, which are near-end signals. Furthermore, the dual microphone array **13** comprises two microphones, namely, a first microphone mic and a second microphone mic². The dual microphone array **13** is an end-firing arrangement. As shown in FIG. 3, the dual microphone array **13** may suppress the sound in the non-gain direction, i.e., the sound in the direction of the speaker **12**, by adopting the end-firing arrangement, so as to retain the sound in the desired gain direction, that is, the sound in the speaking direction of the user at the near end. The controller **11** is configured to acquire the far-end signals generated by the user during the phone conversation, and the first near-end signal of the first microphone mic and the second near-end signal of the second microphone mic².

(15) The electronic equipment **100** further comprises a linear adaptive filter module **14**, an adaptive zero-pole echo canceller **15**, a single-channel noise reduction module NR and a suppression gain filtering output module **16**, and the controller **11** is connected with the linear adaptive filter module **14**, the adaptive zero-pole echo canceller **15**, the single-channel noise reduction module NR and the suppression gain filtering output module **16**.

(16) The linear adaptive filter module **14** is configured to perform linear filtering processing on the far-end signals, the first near-end signal and the second near-end signal to obtain a first initial error frequency spectrum and a second initial error frequency spectrum, and the controller **11** acquires the first initial error frequency spectrum and the second initial error frequency spectrum from the linear adaptive filter module **14**.

(17) The linear adaptive filter module **14** is further configured to perform centroid fitting on the first initial error frequency spectrum and the second initial error frequency spectrum respectively to obtain initial suppression gain factors of the first near-end signal and the second near-end signal, and use the initial suppression gain factors as variable step-size factors for linear filtering processing of the next frame, and the controller **11** obtains the first error frequency spectrum and the first initial suppression gain factor as well as the second error frequency spectrum and the second initial suppression gain factor from the linear adaptive filter module **14**, and the first initial suppression gain factor and the second initial suppression gain factor constitute the initial suppression gain factors.

(18) The adaptive zero-pole echo canceller **15** is configured to perform residual echo suppression on the first error frequency spectrum and the second error frequency spectrum to obtain a target frequency spectrum signal.

(19) The suppression gain filtering output module **16** is configured to obtain a smooth factor and perform full-band smoothing and exponential operation to obtain a secondary suppression gain factor after performing sub-band range selection on the initial suppression gain factor; and perform filtering processing on the target frequency spectrum signal by using the secondary suppression gain factor to obtain a target near-end voice signal, and output the target near-end voice signal.

(20) The electronic equipment **100** further comprises a time delay compensation operation module d/c, which is configured to calculate the time delay compensation between the far-end signal and each of the first near-end signal and/or the second near-end signal respectively.

(21) By performing the preliminary filtering processing, centroid fitting, residual echo suppression and secondary filtering processing on the far-end signals and the near-end signals by using the controller **11**, the linear adaptive filter module **14**, the adaptive zero-pole echo canceller **15** and the suppression gain filtering output module **16**, the electronic equipment **100** realizes the echo cancellation during the phone conversation, solves the problems of poor echo cancellation effect and large residual echo, and effectively improves the quality of voice call.

(22) Please refer to FIG. 4, which is a schematic flowchart diagram of dual-microphone array echo cancellation method provided according to an embodiment of the present invention. The method is

applied to the electronic equipment **100** and may be executed by a processor **111** in the electronic equipment **100**. As shown in FIG. 4, the method comprises: **S401**: acquiring far-end signals generated by an electronic equipment during a phone conversation, and a first near-end signal and a second near-end signal of a dual microphone array.

(23) A controller of the electronic equipment acquires the far-end signals and the near-end signals generated by the electronic equipment during a phone conversation, wherein the far-end signals are sound signals played by the speaker, and the near-end signals include the voice of the user, echo and/or background noise. Furthermore, the dual microphone array comprises a first microphone mic1 and a second microphone mic2, the first microphone generates the first near-end signal and the second microphone generates the second near-end signal.

(24) The far-end signal is represented by $x(n)$, and the far-end signal $x(n)$ is represented by Equation 1:

$$x(n)=[x(i),x(i-1),\dots,x(i-M+1)].sup.T \quad \text{Equation 1;}$$

(25) The first near-end signal is represented by $d1(n)$, and the first near-end signal $d1(n)$ is represented by Equation 2:

$$d1(n)=[d1(i),d1(i-1),\dots,d1(i-M+1)].sup.T \quad \text{Equation 2;}$$

(26) The second near-end signal is represented by $d2(n)$, and the second near-end signal $d2(n)$ is represented by Equation 3:

$$d2(n)=[d2(i),d2(i-1),\dots,d2(i-M+1)].sup.T \quad \text{Equation 3;}$$

(27) wherein M represents the number of sampling points of the far-end signal $x(n)$ or the first near-end signal $d1(n)$ and the second near-end signal $d2(n)$ in a frame, and i represents the i th sampling point of the controller of the electronic equipment when collecting the far-end signal $x(n)$ or the near-end signal $d1(n)$ and the second near-end signal $d2(n)$.

(28) **S402**: performing linear filtering processing on the far-end signals, the first near-end signal and the second near-end signal to obtain a first initial error frequency spectrum and a second initial error frequency spectrum.

(29) After the controller of the electronic equipment obtains the far-end signal $x(n)$, the first near-end signal $d1(n)$ and the second near-end signal $d2(n)$, the linear adaptive filter module performs linear filtering processing on the far-end signal $x(n)$, the first near-end signal $d1(n)$ and the second near-end signal $d2(n)$ to obtain the first initial error frequency spectrum and the second initial error frequency spectrum.

(30) In some embodiments, the operation of performing linear filtering processing on the far-end signals, the first near-end signal and the second near-end signal to obtain a first initial error frequency spectrum and a second initial error frequency spectrum comprises: performing Fourier transform on the far-end signals to obtain far-end frequency domain information, performing Fourier transform on the first near-end signal to obtain first near-end frequency domain information, and performing Fourier transform on the second near-end signal to obtain second near-end frequency domain information; performing filtering processing on the far-end frequency domain information by using the updated filter weight coefficient of the previous frame to obtain an echo frequency spectrum; subtracting the echo frequency spectrum from the first near-end frequency domain information to obtain a first initial error frequency spectrum; subtracting the echo frequency spectrum from the second near-end frequency domain information to obtain a second initial error frequency spectrum.

(31) Specifically, first, Fourier transform is performed on the far-end signal $x(n)$ to obtain the far-end frequency domain information $X(n)$, Fourier transform is performed on the first near-end signal $d1(n)$ to obtain the first near-end frequency domain information $D1(n)$, and then Fourier transform is performed on the second near-end signal $d2(n)$ to obtain the second near-end frequency domain information $D2(n)$. The far-end frequency domain information $X(n)$ is represented by Equation 4, the first near-end frequency domain information $D1(n)$ is represented by Equation 5, and the second near-end frequency domain information $D2(n)$ is represented by Equation 6:

$$X(n)=fft([x(n-1);x(n)].\text{Math.win}) \quad \text{Equation 4;}$$

$$D1(n)=fft([d1(n-1);d1(n)].\text{Math.win}) \quad \text{Equation 5;}$$

$$D2(n)=fft([d2(n-1);d2(n)].\text{Math.win}) \quad \text{Equation 6;}$$

(32) wherein $X(n)$ represents the far-end frequency domain information of the n th frame, $D1(n)$ represents the first near-end frequency domain information of the n th frame, $D2(n)$ represents the second near-end frequency domain information of the n th frame, win represents the Hanning window with a length of $2*M$, and fft represents the Fourier transform. The near-end frequency domain information $D(n)$ includes the first near-end frequency domain information $D1(n)$ and the second near-end frequency domain information $D2(n)$.

(33) Then, filtering processing is performed on the far-end frequency domain information $X(n)$ by using the updated filter weight coefficient of the previous frame to obtain the echo frequency spectrum. Because the echo path and the process of phone communication are fixed, the far-end frequency domain information may be subjected to delay alignment respectively with the first near-end frequency domain information $D1(n)$ and the second near-end frequency domain information $D2(n)$ before filtering, so as to estimate the first echo frequency spectrum $Y1(n)$ and the second echo frequency spectrum $Y2(n)$. The updated filter weight coefficient of the previous frame includes the updated filter weight coefficient $W.\text{sub}.1(n-1)$ of the previous frame for the first microphone and the updated filter weight coefficient $W.\text{sub}.2(n-1)$ of the previous frame for the second microphone. That is, the far-end frequency domain information $X(n)$ is filtered in the frequency domain, the obtained echo frequency spectrum is represented by $Y(n)$, and the echo frequency spectrum $Y(n)$ includes the first echo frequency spectrum $Y1(n)$ and the second echo frequency spectrum $Y2(n)$ which are calculated by using Equation 7 and Equation 8:

$$Y1(n)=W.\text{sub}.1(n-1).\text{Math}.X(n) \quad \text{Equation 7;}$$

$$Y2(n)=W.\text{sub}.2(n-1).\text{Math}.X(n) \quad \text{Equation 8.}$$

(34) wherein $W.\text{sub}.1(n-1)$ represents the updated filter weight coefficient of the previous frame for the first microphone, and $W.\text{sub}.2(n-1)$ represents the updated filter weight coefficient of the previous frame for the second microphone.

(35) Finally, the echo frequency spectrum $Y(n)$ is subtracted from the near-end frequency domain information $D(n)$ to obtain the first initial error frequency spectrum $E1(n)$ and the second initial error frequency spectrum $E2(n)$. That is, the first echo frequency spectrum $Y1(n)$ is subtracted from the first near-end frequency domain information $D1(n)$ to obtain the first initial error frequency spectrum $E1(n)$; and the second echo frequency spectrum $Y2(n)$ is subtracted from the second near-end frequency domain information $D2(n)$ to obtain the second initial error frequency spectrum $E2(n)$. $E1(n)$ and $E2(n)$ are represented by Equation 9 and Equation 10:

$$E1(n)=D1(n-1)-Y1(n) \quad \text{Equation 9;}$$

$$E2(n)=D2(n-1)-Y2(n) \quad \text{Equation 10.}$$

(36) The first initial error frequency spectrum $E1(n)$ and the second initial error frequency spectrum $E2(n)$ that are obtained can preliminarily eliminate the linear echo of the first microphone and the second microphone.

(37) **S403:** performing centroid fitting on the first initial error frequency spectrum and the second initial error frequency spectrum respectively to obtain initial suppression gain factors of the first near-end signal and the second near-end signal, and using the initial suppression gain factors as variable step-size factors for linear filtering processing of the next frame.

(38) In order to reduce most of the linear echo of the near-end signal, first it is necessary to calculate the initial suppression gain factors, and the initial suppression gain factors include the first initial suppression gain factor and the second initial suppression gain factor. The operation of performing centroid fitting on the first initial error frequency spectrum and the second initial error frequency spectrum respectively to obtain initial suppression gain factors of the first near-end signal and the second near-end signal may comprise: calculating a first cross-correlation coefficient between the far-end frequency domain information and the first initial error frequency spectrum,

and a second cross-correlation coefficient between the far-end frequency domain information and the second initial error frequency spectrum; performing centroid fitting on the first cross-correlation coefficient to obtain a first initial suppression gain factor, and performing centroid fitting on the second cross-correlation coefficient to obtain a second initial suppression gain factor; multiplying the first initial suppression gain factor by the first initial error frequency spectrum to obtain the first error frequency spectrum, and multiplying the second initial suppression gain factor by the second initial error frequency spectrum to obtain the second error frequency spectrum.

(39) Specifically, first, the first cross-correlation coefficient between the far-end frequency domain information $X(n)$ and the first initial error frequency spectrum $E1(n)$ is calculated. Similarly, the second cross-correlation coefficient between the far-end frequency domain information $X(n)$ and the second initial error frequency spectrum $E2(n)$ is calculated.

(40) The first cross-correlation coefficient and the second cross-correlation coefficient are calculated by using Equations 11 to 17.

$$S_{sub.e1}(n) = \gamma \cdot S_{sub.e1}(n-1) + (1-\gamma)E1(n) \cdot \text{Math.conj}(E1(n)) \quad \text{Equation 11;}$$

$$S_{sub.e2}(n) = \gamma \cdot S_{sub.e2}(n-1) + (1-\gamma)E2(n) \cdot \text{Math.conj}(E2(n)) \quad \text{Equation 12;}$$

$$S_{sub.x}(n) = \gamma \cdot S_{sub.x}(n-1) + (1-\gamma)X(n) \cdot \text{Math.conj}(X(n)) \quad \text{Equation 13;}$$

$$S_{sub.xe1}(n) = \gamma \cdot S_{sub.xe1}(n-1) + (1-\gamma)S_{sub.x}(n) \cdot \text{Math.conj}(S_{sub.xe1}(n)) \quad \text{Equation 14;}$$

$$S_{sub.xe2}(n) = \gamma \cdot S_{sub.xe2}(n-1) + (1-\gamma)S_{sub.x}(n) \cdot \text{Math.conj}(S_{sub.xe2}(n)) \quad \text{Equation 15;}$$

$$Coh_{sub.xe1}(n) = S_{sub.xe1}(n) \cdot \text{Math.conj}(S_{sub.xe1}(n)) / (S_{sub.x}(n) \cdot \text{Math.S}_{sub.e1}(n) + \sigma) \quad \text{Equation 16;}$$

$$Coh_{sub.xe2}(n) = S_{sub.xe2}(n) \cdot \text{Math.conj}(S_{sub.xe2}(n)) / (S_{sub.x}(n) \cdot \text{Math.S}_{sub.e2}(n) + \sigma) \quad \text{Equation 17.}$$

wherein γ represents the smooth factor, of which the value is generally a number less than 1 but close to 1, e.g., 0.9; σ represents the division protection factor, which may be set $\sigma > 0$; conj represents conjugate operation; $S_{sub.e1}(n)$ represents a smooth power spectrum of the first initial error frequency spectrum; $S_{sub.e2}(n)$ represents the smooth power spectrum of the second initial error frequency spectrum; $S_{sub.x}(n)$ represents the smooth power spectrum of the far-end frequency domain information, $S_{sub.xe1}(n)$ represents the smooth power spectrum of the far-end frequency domain information, the first near-end frequency domain information and the first initial error frequency spectrum, $S_{sub.xe2}(n)$ represents the smooth power spectrum of the far-end frequency domain information, the second near-end frequency domain information and the second initial error frequency spectrum; $Coh_{sub.xe1}(n)$ represents the first cross-correlation coefficient, and $Coh_{sub.xe2}(n)$ represents the second cross-correlation coefficient.

(41) Then, centroid fitting is performed on the first cross-correlation coefficient to obtain the first initial suppression gain factor, and centroid fitting is performed on the second cross-correlation coefficient to obtain the second initial suppression gain factor, and the first initial suppression gain factor and the second initial suppression gain factor may be calculated by using Equation 18 and Equation 19:

$$(42) \quad g1(i) = 1 - \frac{\text{Math.} \sum_{j=i-k}^{i+k} Coh_{xe1}(j)}{\text{Math.} \sum_{j=i-k}^{i+k} Coh_{xe2}(j)}; \quad \text{Equation 18} \quad g2(i) = 1 - \frac{\text{Math.} \sum_{j=i-k}^{i+k} Coh_{xe1}(j)}{\text{Math.} \sum_{j=i-k}^{i+k} Coh_{xe2}(j)}; \quad \text{Equation 19}$$

wherein k represents the centroid fitting distance, i represents the i th frequency point of the n th frame, and $g1(i)$ represents the first initial suppression gain factor; and $g2(i)$ represents the second initial suppression gain factor.

(43) After the first initial suppression gain factor $g1(i)$ and the second initial suppression gain factor $g2(i)$ are obtained, the first initial suppression gain factor $g1(i)$ is multiplied by the first initial error frequency spectrum $E1(n)$ to obtain the first error frequency spectrum, and the second initial suppression gain factor $g2(i)$ is multiplied by the second initial error frequency spectrum $E2(n)$ to obtain the second error frequency spectrum, and the first error frequency spectrum and the second

error frequency spectrum are represented by Equation 20 and Equation 21:

$$E1'(n)=E1(n).\text{Math}.g1(n) \quad \text{Equation 20};$$

$$E2'(n)=E2(n).\text{Math}.g2(n) \quad \text{Equation 21}.$$

(44) wherein $E1'(n)$ represents the first error frequency spectrum, and $E2'(n)$ represents the second error frequency spectrum.

(45) The first error frequency spectrum $E1'(n)$ and the second error frequency spectrum $E2'(n)$ that are obtained can reduce the near-end frequency domain signals of most linear echoes.

(46) In some embodiments, in order to improve the accuracy and robustness of the filter, the filter weight coefficient may be updated in real time. Therefore, the operation of using the initial suppression gain factors as variable step-size factors for linear filtering processing of the next frame may comprise: updating the filter weight coefficient of the previous frame based on the first initial suppression gain factor and the second initial suppression gain factor to obtain the filter weight coefficient of the current frame; using the first initial suppression gain factor and the second initial suppression gain factor as variable step-size factors for linear filtering processing of the next frame, if the first initial suppression gain factor and the second initial suppression gain factor are smaller than a preset threshold.

(47) Specifically, the filter weight coefficient $W(n-1)$ of the previous frame is updated based on the first initial suppression gain factor $g1(i)$ and the second initial suppression gain factor $g2(i)$ to obtain the filter weight coefficient of the current frame, wherein the filter weight coefficient of the previous frame includes the updated filter weight coefficient $W.\text{sub}.1(n-1)$ of the previous frame for the first microphone and the updated filter weight coefficient $W.\text{sub}.2(n-1)$ of the previous frame for the second microphone, and the filter weight coefficients of the current frame are represented by Equation 22 and Equation 23:

$$W.\text{sub}.1(n)=W.\text{sub}.1(n-1)+\mu.\text{Math}.g1(n).\text{Math}.\Delta W.\text{sub}.1 \quad \text{Equation 22};$$

$$W.\text{sub}.2(n)=W.\text{sub}.2(n-1)+\mu.\text{Math}.g2(n).\text{Math}.\Delta W.\text{sub}.2 \quad \text{Equation 23}.$$

(48) wherein $W.\text{sub}.1(n)$ represents the filter weight coefficient of the current frame for the first microphone mic and $W.\text{sub}.2(n)$ represents the filter weight coefficient of the current frame for the second microphone mic2; μ represents a fixed step factor, of which the value is usually in the range of $0<\mu<2$; $\Delta W.\text{sub}.1$ represents the adjustment amount of the filter coefficient of the first microphone mic1, and $\Delta W.\text{sub}.2$ represents the adjustment amount of the filter coefficient of the second microphone mic2. The calculation methods of the adjustment amount of the filter coefficient include but not limited to normalized least mean square error (NLMS), recursive least squares (RLS) and Kalman algorithm or the like; $g1(n)$ represents the first initial suppression gain factor, and $g2(n)$ represents the second initial suppression gain factor. $g1(n)$ and $g2(n)$ are used as variable step-size factors for linear filtering processing of the next frame, and the step of performing filtering processing on the far-end frequency domain information is executed so that the filter is more robust.

(49) Therefore, when the filter does not converge, the residual echo is too large and the step size becomes larger, which helps the filter to converge quickly. When the filter gradually converges, for example, when the first initial suppression gain factor $g1(n)$ and the second initial suppression gain factor $g2(n)$ gradually become smaller to be less than a preset threshold and close to 0 (for example, the preset threshold is 0.01), the step size gradually decreases, which indicates that the residual echo in the first initial error frequency spectrum $E1(n)$ and the second initial error frequency spectrum $E2(n)$ is small or in an echoless state. Then, according to step S402, filtering processing is performed on the far-end frequency domain information $X(n)$ using the filter weight coefficients ($W.\text{sub}.1(n)$ and $W.\text{sub}.2(n)$) of the current frame to obtain the echo frequency spectrum, and then the initial error frequency spectrum is calculated.

(50) S404: performing residual echo suppression on the first error frequency spectrum and the second error frequency spectrum by using an adaptive zero-pole echo canceller to obtain a target frequency spectrum signal.

(51) In some embodiments, the operation of performing residual echo suppression on the first error frequency spectrum and the second error frequency spectrum by using an adaptive zero-pole echo canceller to obtain a target frequency spectrum signal may comprise: calculating a first branch frequency spectrum signal based on the first error frequency spectrum and a time delay compensation, and calculating a second branch frequency spectrum signal based on the second error frequency spectrum and the time delay compensation; calculating a differential signal cross-correlation power spectrum coefficient between the first branch frequency spectrum signal and the second branch frequency spectrum signal and a weight coefficient of the adaptive zero-pole echo canceller based on the first branch frequency spectrum signal and the second branch frequency spectrum signal; performing residual echo suppression on the first error frequency spectrum and the second error frequency spectrum by using the weight coefficient of the adaptive zero-pole echo canceller to obtain the target frequency spectrum signal.

(52) Specifically, first, the frequency spectrum signals of two branches are calculated. That is, the first branch frequency spectrum signal is calculated based on the first error frequency spectrum and the time delay compensation, and the second branch frequency spectrum signal is calculated based on the second error frequency spectrum and the time delay compensation. Then, a differential signal cross-correlation power spectrum coefficient between the first branch frequency spectrum signal and the second branch frequency spectrum signal as well as a weight coefficient of the adaptive zero-pole echo canceller are calculated based on the first branch frequency spectrum signal and the second branch frequency spectrum signal, and finally residual echo suppression is performed on the first error frequency spectrum and the second error frequency spectrum by using the weight coefficient of the adaptive zero-pole echo canceller to obtain the target frequency spectrum signal. These operations may be performed by using Equation 24 to Equation 29:

$$F(n)=E2(n)-\text{delay}(E1(n)) \quad \text{Equation 24;}$$

$$B(n)=\text{delay}(E2(n))-E1(n) \quad \text{Equation 25;}$$

$$R.\text{sub}.FB(n)=\alpha F(n)B(n)+\beta R.\text{sub}.FB(n-1) \quad \text{Equation 26;}$$

$$R.\text{sub}.BB(n)=\alpha B.\text{sup}.2(n)+\beta R.\text{sub}.BB(n-1) \quad \text{Equation 27;}$$

$$W.\text{sub}.3(n)=R.\text{sub}.FB(n)/R.\text{sub}.BB(n) \quad \text{Equation 28;}$$

$$E(n)=F(n)-W.\text{sub}.3(n).\text{Math}.B(n) \quad \text{Equation 29.}$$

wherein $F(n)$ represents the second branch frequency spectrum signal; $B(n)$ represents the first branch frequency spectrum signal; delay represents time delay compensation operation, $\text{delay}(E1(n))$ represents the time delay compensation of the first branch, and $\text{delay}(E2(n))$ represents the time delay compensation of the second branch; the range of the value of α is $0 \leq \alpha \leq 1$, the range of the value of β is $0 \leq \beta \leq 1$, and $\alpha + \beta = 1$; $W.\text{sub}.3(n)$ represents the weight coefficient of the adaptive zero-pole echo canceller, $R.\text{sub}.FB(n)$ represents the differential signal cross-correlation power spectrum coefficient, $R.\text{sub}.BB(n)$ represents the second branch autocorrelation power spectrum; and $E(n)$ represents the target frequency spectrum signal.

(53) **S405:** after performing sub-band range selection on the initial suppression gain factors, obtaining a smooth factor, and performing full-band smoothing and exponential operation to obtain a secondary suppression gain factor.

(54) In order to improve the echo cancellation effect, smoothing may be performed on the initial suppression gain factor.

(55) In some embodiments, the initial suppression gain factors include the first initial suppression gain factor and the second initial suppression gain factor, the operation of after performing sub-band range selection on the initial suppression gain factors, obtaining a smooth factor, and performing full-band smoothing and exponential operation to obtain a secondary suppression gain factor may comprise: performing sub-band range selection on the initial suppression gain factors to obtain a plurality of sub-band ranges; sorting the initial suppression gain factors in the plurality of sub-band ranges, and calculating a suppression gain coefficient smooth factor; performing full-band smoothing processing and exponential operation on the initial suppression gain factor by

using the suppression gain coefficient smooth factor to obtain the secondary suppression gain factor.

(56) Specifically, the sub-band ranges may be selected by selecting the sub-band minimum minFre, the sub-band maximum maxFre and the reference sub-band point refFre. Then, the sub-band ranges are sorted for example in an ascending order to obtain the sequence hNI, and the reference value Refer is selected according to Equation 30, which is as follows:

$$\text{Refer} = \text{hNI}[\text{refFre}] \quad \text{Equation 30.}$$

(57) wherein Refer represents the reference value; refFre represents the reference sub-band point; hNI represents the sequence of the sub-band range in the ascending order.

(58) The suppression gain coefficient smooth factor may be calculated by using Equation 31, which is as follows:

$$\text{Smooth}(n) = 1 / (\max \text{Fre} - \min \text{fre}) \cdot \sum_{i=\min \text{Fre}}^{\max \text{Fre}} \text{hNI}[i] \quad \text{Equation 31.}$$

(59) wherein Smooth(n) represents the suppression gain coefficient smooth factor; minFre represents the sub-band minimum; maxFre represents the sub-band maximum, and refFre represents the reference sub-band point; i represents the ith frequency point of the nth frame.

(60) After obtaining the suppression gain coefficient smooth factor Smooth(n), full-band smoothing processing and exponential operation are performed on the initial suppression gain factor by using the suppression gain coefficient smooth factor Smooth(n) to obtain the secondary suppression gain factor, which is represented by Equation 32:

$$G(i) = \text{Smooth}(n) \cdot \text{Refer} + (1 - \text{Smooth}(n)) / (0.5 \cdot \text{Math.g1}(i) + 0.5 \cdot \text{Math.g2}(i)) \quad \text{Equation 32;}$$

(61) wherein the initial suppression gain factor includes the first initial suppression gain factor g1(i) and the second initial suppression gain factor g2(i), and G(i) represents the secondary suppression gain factor.

(62) **S406**: performing filtering processing on the target frequency spectrum signal by using the secondary suppression gain factor to obtain a target near-end voice signal.

(63) In order to cancel the residual echo as much as possible, and at the same time to avoid the possible damage to the near-end signal by the secondary filtering so as to obtain the optimal output, the operation of performing filtering processing on the target frequency spectrum signal by using the secondary suppression gain factor to obtain a target near-end voice signal may comprise: performing noise reduction processing on the target frequency spectrum signal; multiplying the secondary suppression gain factor by the target frequency spectrum signal after noise reduction, then performing inverse Fourier transform, and performing overlap-adding to obtain the target near-end voice signal.

(64) Specifically, first, noise reduction processing is performed on the target frequency spectrum signal, then after the secondary suppression gain factor is multiplied by the target frequency spectrum signal after noise reduction, inverse Fourier transform is performed, and overlap-adding is performed to obtain the target near-end voice signal. These operations are performed according to Equation 33 to Equation 36:

$$E.\text{sub.nr}(n) = \text{NR}(E'(n)) \quad \text{Equation 33;}$$

$$e(n) = \text{ifft}(E.\text{sub.nr}(n) \cdot \text{Math.G}(n)) \quad \text{Equation 34;}$$

$$\text{out}(n) = e(1:M) + \text{ola_buf} \quad \text{Equation 35;}$$

$\text{ola_buf} = e(M+1:2 \cdot M)$ Equation 36. wherein ola_buf represents the overlap-adding reserved block, ifft represents inverse Fourier transform; NR represents noise reduction algorithm processing, E.sub.nr(n) represents the target frequency spectrum signal after noise reduction processing, E'(n) represents the error frequency spectrum, which includes the first error frequency spectrum E1'(n) and the second error frequency spectrum E2'(n); out(n) represents the target near-end voice signal.

(65) According to the embodiment of the present invention, far-end signals generated by an electronic equipment during a phone conversation as well as a first near-end signal and a second near-end signal of a dual microphone array are acquired, then preliminary filtering processing is

performed on the far-end signals and the near-end signals of the dual microphone array, e.g., linear filtering processing may be performed on the far-end signals, the first near-end signal and the second near-end signal to obtain a first initial error frequency spectrum and a second initial error frequency spectrum so as to preliminarily cancel the echo; next, centroid fitting is performed on the first initial error frequency spectrum and the second initial error frequency spectrum respectively to obtain initial suppression gain factors of the first near-end signal and the second near-end signal, and the initial suppression gain factors are used as variable step-size factors for linear filtering processing of the next frame; residual echo suppression is performed on the first error frequency spectrum and the second error frequency spectrum by using an adaptive zero-pole echo canceller to obtain a target frequency spectrum signal so as to achieve residual echo suppression; after performing sub-band range selection on the initial suppression gain factor, a smooth factor is obtained, and full-band smoothing and exponential operation are performed to obtain a secondary suppression gain factor; filtering processing is performed on the target frequency spectrum signal by using the secondary suppression gain factor to achieve secondary suppression gain filtering and obtain a target near-end voice signal, thereby effectively improving the echo cancellation effect and improving the quality of voice call.

(66) Correspondingly, as shown in FIG. 5, an embodiment of the present invention further provides a dual-microphone array echo cancellation device, which may be used in an electronic equipment. The dual-microphone array echo cancellation device **500** comprises: a signal acquisition module **501**, being configured to acquire far-end signals generated by an electronic equipment during a phone conversation, and a first near-end signal and a second near-end signal of the dual microphone array; a linear filtering module **502**, being configured to perform linear filtering processing on the far-end signals, the first near-end signal and the second near-end signal to obtain a first initial error frequency spectrum and a second initial error frequency spectrum; a centroid fitting module **503**, being configured to perform centroid fitting on the first initial error frequency spectrum and the second initial error frequency spectrum respectively to obtain initial suppression gain factors of the first near-end signal and the second near-end signal, and use the initial suppression gain factors as variable step-size factors for linear filtering processing of the next frame; a residual echo suppression module **504**, being configured to perform residual echo suppression on the first error frequency spectrum and the second error frequency spectrum by using an adaptive zero-pole echo canceller to obtain a target frequency spectrum signal; a smoothing module **505**, being configured to obtain a smooth factor and perform full-band smoothing and exponential operation to obtain a secondary suppression gain factor after performing sub-band range selection on the initial suppression gain factor; a target acquisition module **506**, being configured to perform filtering processing on the target frequency spectrum signal by using the secondary suppression gain factor to obtain a target near-end voice signal.

(67) According to the embodiment of the present invention, far-end signals generated by an electronic equipment during a phone conversation as well as a first near-end signal and a second near-end signal of a dual microphone array are acquired, then preliminary filtering processing is performed on the far-end signals and the near-end signals of the dual microphone array, e.g., linear filtering processing may be performed on the far-end signals, the first near-end signal and the second near-end signal to obtain a first initial error frequency spectrum and a second initial error frequency spectrum so as to preliminarily cancel the echo; next, centroid fitting is performed on the first initial error frequency spectrum and the second initial error frequency spectrum respectively to obtain initial suppression gain factors of the first near-end signal and the second near-end signal, and the initial suppression gain factors are used as variable step-size factors for linear filtering processing of the next frame; residual echo suppression is performed on the first error frequency spectrum and the second error frequency spectrum by using an adaptive zero-pole echo canceller to obtain a target frequency spectrum signal so as to achieve residual echo suppression; after performing sub-band range selection on the initial suppression gain factor, a smooth factor is

obtained, and full-band smoothing and exponential operation are performed to obtain a secondary suppression gain factor; filtering processing is performed on the target frequency spectrum signal by using the secondary suppression gain factor to achieve secondary suppression gain filtering and obtain a target near-end voice signal, thereby effectively improving the echo cancellation effect and improving the quality of voice call.

(68) In other embodiments, the linear filtering module **502** is further configured to: perform Fourier transform on the far-end signals to obtain far-end frequency domain information, performing Fourier transform on the first near-end signal to obtain first near-end frequency domain information, and performing Fourier transform on the second near-end signal to obtain second near-end frequency domain information; perform filtering processing on the far-end frequency domain information by using the updated filter weight coefficient of the previous frame to obtain an echo frequency spectrum; subtract the echo frequency spectrum from the first near-end frequency domain information to obtain a first initial error frequency spectrum; subtract the echo frequency spectrum from the second near-end frequency domain information to obtain a second initial error frequency spectrum.

(69) In other embodiments, the centroid fitting module **503** is further configured to: calculate a first cross-correlation coefficient between the far-end frequency domain information and the first initial error frequency spectrum, and a second cross-correlation coefficient between the far-end frequency domain information and the second initial error frequency spectrum; perform centroid fitting on the first cross-correlation coefficient to obtain a first initial suppression gain factor, and perform centroid fitting on the second cross-correlation coefficient to obtain a second initial suppression gain factor; multiply the first initial suppression gain factor by the first initial error frequency spectrum to obtain the first error frequency spectrum, and multiply the second initial suppression gain factor by the second initial error frequency spectrum to obtain the second error frequency spectrum.

(70) In other embodiments, the echo cancellation device **500** for a dual microphone array further comprises an updating module **507**, which is configured to: update the filter weight coefficient of the previous frame based on the first initial suppression gain factor and the second initial suppression gain factor to obtain the filter weight coefficient of the current frame; use the first initial suppression gain factor and the second initial suppression gain factor as variable step-size factors for linear filtering processing of the next frame, if the first initial suppression gain factor and the second initial suppression gain factor are smaller than a preset threshold.

(71) In other embodiments, the residual echo suppression module **504** is further configured to: calculate a first branch frequency spectrum signal based on the first error frequency spectrum and a time delay compensation, and calculating a second branch frequency spectrum signal based on the second error frequency spectrum and the time delay compensation; calculate a differential signal cross-correlation power spectrum coefficient between the first branch frequency spectrum signal and the second branch frequency spectrum signal and a weight coefficient of the adaptive zero-pole echo canceller based on the first branch frequency spectrum signal and the second branch frequency spectrum signal; perform residual echo suppression on the first error frequency spectrum and the second error frequency spectrum by using the weight coefficient of the adaptive zero-pole echo canceller to obtain the target frequency spectrum signal.

(72) In other embodiments, the smoothing module **505** is further configured to: perform sub-band range selection on the initial suppression gain factors to obtain the initial suppression gain factors in a plurality of sub-band ranges; sort the initial suppression gain factors in the plurality of sub-band ranges, and calculate a suppression gain coefficient smooth factor; perform full-band smoothing processing and exponential operation on the initial suppression gain factor by using the suppression gain coefficient smooth factor to obtain the secondary suppression gain factor.

(73) In other embodiments, the target acquisition module **506** is further configured to: perform noise reduction processing on the target frequency spectrum signal; multiply the secondary

suppression gain factor by the target frequency spectrum signal after noise reduction, perform inverse Fourier transform, and perform overlap-adding to obtain the target near-end voice signal. (74) It shall be noted that, the above device may execute the method provided according to the embodiment of the present application, and has corresponding functional modules and beneficial effects for executing the method. Reference may be made to the method provided according to the embodiment of the present application for technical details not described in detail in the embodiment of the device.

(75) FIG. 6 is a schematic view of the hardware structure of a controller of the electronic equipment **100** in one embodiment of the electronic equipment **100**. As shown in FIG. 6, the controller **110** comprises: one or more processors **111** and a memory **112**. In FIG. 6, one processor **111** and one memory **112** are taken as examples.

(76) The processor **111** and the memory **112** may be connected by a bus or other means, and the bus connection is taken as an example in FIG. 6.

(77) As a nonvolatile computer readable storage medium, the memory **112** may be used to store nonvolatile software programs, nonvolatile computer executable programs and modules, such as program instructions/modules (e.g., the signal acquisition module **501**, the linear filtering module **502**, the centroid fitting module **503**, the residual echo suppression module **504**, the smoothing module **505**, the target acquisition module **506** and the updating module **507** shown in FIG. 5) corresponding to the dual-microphone array echo cancellation method in the embodiment of the present application. The processor **111** executes various functional applications and data processing of the controller, i.e., implements the dual-microphone array echo cancellation method provided by the above embodiments of the method, by running the nonvolatile software programs, instructions and modules stored in the memory **112**.

(78) The memory **112** may comprise a program storage area and a data storage area, wherein the program storage area may store operating systems and application programs required by at least one function; and the data storage area may store data created according to the use of the electronic equipment **100** or the like. In addition, the memory **112** may comprise a high-speed random access memory, and may also comprise a nonvolatile memory, such as at least one magnetic disk memory device, flash memory device, or other nonvolatile solid-state memory devices. In some embodiments, the memory **112** optionally comprises memories remotely provided relative to the processor **111**, and these remote memories may be connected to a signal long-time recording equipment through a network. Examples of the above network comprise, but not limited to, the Internet, Intranet, local area networks, mobile communication networks and combinations thereof.

(79) The one or more modules are stored in the memory **112**, and when executed by the one or more processors **111**, the one or more modules execute the dual-microphone array echo cancellation method in any of the embodiments of the method described above, e.g., execute the above-described method steps of **S401** to **S406** of the steps of the method in FIG. 4; and implement the functions of the modules **501** to **507** in FIG. 5.

(80) The products described above may execute the method provided according to the embodiments of the present application, and have corresponding functional modules and beneficial effects for executing the method. For technical details not described in detail in this embodiment, reference may be made to the method provided according to the embodiments of the present application.

(81) An embodiment of the present application provides a nonvolatile computer readable storage medium, in which computer executable instructions are stored. The computer executable instructions, when executed by one or more processors, e.g., one processor **111** in FIG. 6, may enable the one or more processors described above to execute the dual-microphone array echo cancellation method in any of the embodiments of the method described above, e.g., execute the above-described method steps of **S401** to **S406** of the steps of the method in FIG. 4; and implement the functions of the modules **501** to **507** in FIG. 5.

(82) The embodiments of the devices described above are only for illustrative purpose. The units illustrated as separate components may be or may not be physically separated, and components displayed as units may be or may not be physical units. That is, these units and components may be located in one place or distributed over multiple network units. Some or all of the modules may be selected according to actual needs to achieve the purpose of the solution of the embodiment.

(83) From the description of the above embodiments, those of ordinary skill in the art may clearly appreciate that each embodiment may be realized by means of software plus a general hardware platform, and of course, it may also be realized by hardware. As shall be appreciated by those of ordinary skill in the art, the implementation of all or part of the processes in the embodiments of the methods described above may be completed by instructing related hardware through a computer program, and the program may be stored in a computer readable storage medium. When it is executed, the program may comprise the processes of the embodiments of the methods described above. The storage medium may be a magnetic disk, an optical disk, a Read-Only Memory (ROM) or a Random Access Memory (RAM) or the like.

(84) Finally, it shall be noted that, the above embodiments are only used to illustrate the technical solutions of the present invention, and are not intended to limit the present invention. Under the idea of the present invention, technical features in the above embodiments or different embodiments may also be combined, the steps may be implemented in any order, and many other variations in different aspects of the present invention as described above are possible, and these variations are not provided in details for conciseness. Although the present invention has been described in detail with reference to the foregoing embodiments, those of ordinary skill in the art shall appreciate that, the technical solutions described in the foregoing embodiments may still be modified or some of the technical features may be equivalently replaced. These modifications or replacements do not make the essence of the corresponding technical solutions deviate from the scope of the technical solutions of various embodiments of the present invention.

Claims

1. A dual-microphone array echo cancellation method, the method comprising: acquiring far-end signals generated by an electronic equipment during a phone conversation, and a first near-end signal and a second near-end signal of a dual microphone array; performing Fourier transform on the far-end signals to obtain far-end frequency domain information, performing Fourier transform on the first near-end signal to obtain first near-end frequency domain information, and performing Fourier transform on the second near-end signal to obtain second near-end frequency domain information; performing filtering processing on the far-end frequency domain information by using the updated filter weight coefficient of the previous frame to obtain an echo frequency spectrum; subtracting the echo frequency spectrum from the first near-end frequency domain information to obtain a first initial error frequency spectrum; subtracting the echo frequency spectrum from the second near-end frequency domain information to obtain a second initial error frequency spectrum; performing centroid fitting on the first initial error frequency spectrum and the second initial error frequency spectrum respectively to obtain initial suppression gain factors of the first near-end signal and the second near-end signal, which comprises calculating a first cross-correlation coefficient between the far-end frequency domain information and the first initial error frequency spectrum, and a second cross-correlation coefficient between the far-end frequency domain information and the second initial error frequency spectrum, performing centroid fitting on the first cross-correlation coefficient to obtain a first initial suppression gain factor, and performing centroid fitting on the second cross-correlation coefficient to obtain a second initial suppression gain factor, multiplying the first initial suppression gain factor by the first initial error frequency spectrum to obtain the first error frequency spectrum, and multiplying the second initial suppression gain factor by the second initial error frequency spectrum to obtain the second error frequency spectrum, and

using the initial suppression gain factors as variable step-size factors for linear filtering processing of the next frame; performing residual echo suppression on the first error frequency spectrum and the second error frequency spectrum by using an adaptive zero-pole echo canceller to obtain a target frequency spectrum signal; after performing sub-band range selection on the initial suppression gain factors, obtaining a smooth factor, and performing full-band smoothing and exponential operation to obtain a secondary suppression gain factor; performing filtering processing on the target frequency spectrum signal by using the secondary suppression gain factor to obtain a target near-end voice signal.

2. The method of claim 1, the operation of using the initial suppression gain factors as variable step-size factors for linear filtering processing of the next frame comprises: updating the filter weight coefficient of the previous frame based on the first initial suppression gain factor and the second initial suppression gain factor to obtain the filter weight coefficient of the current frame; using the first initial suppression gain factor and the second initial suppression gain factor as variable step-size factors for linear filtering processing of the next frame, if the first initial suppression gain factor and the second initial suppression gain factor are smaller than a preset threshold.

3. The method of claim 1, the operation of performing residual echo suppression on the first error frequency spectrum and the second error frequency spectrum by using an adaptive zero-pole echo canceller to obtain a target frequency spectrum signal comprises: calculating a first branch frequency spectrum signal based on the first error frequency spectrum and a time delay compensation, and calculating a second branch frequency spectrum signal based on the second error frequency spectrum and the time delay compensation; calculating a differential signal cross-correlation power spectrum coefficient between the first branch frequency spectrum signal and the second branch frequency spectrum signal and a weight coefficient of the adaptive zero-pole echo canceller based on the first branch frequency spectrum signal and the second branch frequency spectrum signal; performing residual echo suppression on the first error frequency spectrum and the second error frequency spectrum by using the weight coefficient of the adaptive zero-pole echo canceller to obtain the target frequency spectrum signal.

4. The method of claim 1, the operation of after performing sub-band range selection on the initial suppression gain factors, obtaining a smooth factor, and performing full-band smoothing and exponential operation to obtain a secondary suppression gain factor comprises: performing sub-band range selection on the initial suppression gain factors to obtain the initial suppression gain factors in a plurality of sub-band ranges; sorting the initial suppression gain factors in the plurality of sub-band ranges, and calculating a suppression gain coefficient smooth factor; performing full-band smoothing processing and exponential operation on the initial suppression gain factor by using the suppression gain coefficient smooth factor to obtain the secondary suppression gain factor.

5. The method of claim 1, the operation of performing filtering processing on the target frequency spectrum signal by using the secondary suppression gain factor to obtain a target near-end voice signal comprises: performing noise reduction processing on the target frequency spectrum signal; multiplying the secondary suppression gain factor by the target frequency spectrum signal after noise reduction, performing inverse Fourier transform, and performing overlap-adding to obtain the target near-end voice signal.

6. An electronic equipment, the electronic equipment comprising: a dual microphone array, at least one processor, and a memory, wherein the memory is communicatively connected with the processor, and the memory stores instructions executable by the at least one processor, and the instructions are executed by the at least one processor to enable the at least one processor to perform: acquiring far-end signals generated by an electronic equipment during a phone conversation, and a first near-end signal and a second near-end signal of a dual microphone array; performing Fourier transform on the far-end signals to obtain far-end frequency domain

information, performing Fourier transform on the first near-end signal to obtain first near-end frequency domain information, and performing Fourier transform on the second near-end signal to obtain second near-end frequency domain information; performing filtering processing on the far-end frequency domain information by using the updated filter weight coefficient of the previous frame to obtain an echo frequency spectrum; subtracting the echo frequency spectrum from the first near-end frequency domain information to obtain a first initial error frequency spectrum; subtracting the echo frequency spectrum from the second near-end frequency domain information to obtain a second initial error frequency spectrum; performing centroid fitting on the first initial error frequency spectrum and the second initial error frequency spectrum respectively to obtain initial suppression gain factors of the first near-end signal and the second near-end signal, which comprises calculating a first cross-correlation coefficient between the far-end frequency domain information and the first initial error frequency spectrum, and a second cross-correlation coefficient between the far-end frequency domain information and the second initial error frequency spectrum, performing centroid fitting on the first cross-correlation coefficient to obtain a first initial suppression gain factor, and performing centroid fitting on the second cross-correlation coefficient to obtain a second initial suppression gain factor, multiplying the first initial suppression gain factor by the first initial error frequency spectrum to obtain the first error frequency spectrum, and multiplying the second initial suppression gain factor by the second initial error frequency spectrum to obtain the second error frequency spectrum, and using the initial suppression gain factors as variable step-size factors for linear filtering processing of the next frame; performing residual echo suppression on the first error frequency spectrum and the second error frequency spectrum by using an adaptive zero-pole echo canceller to obtain a target frequency spectrum signal; after performing sub-band range selection on the initial suppression gain factors, obtaining a smooth factor, and performing full-band smoothing and exponential operation to obtain a secondary suppression gain factor; performing filtering processing on the target frequency spectrum signal by using the secondary suppression gain factor to obtain a target near-end voice signal.

7. The electronic equipment of claim 6, the dual microphone array is an end-firing arrangement.

8. A non-transitory computer readable storage medium, the computer readable storage medium storing computer-executable instructions which, when executed by an electronic equipment, cause the electronic equipment to perform: acquiring far-end signals generated by an electronic equipment during a phone conversation, and a first near-end signal and a second near-end signal of a dual microphone array; performing Fourier transform on the far-end signals to obtain far-end frequency domain information, performing Fourier transform on the first near-end signal to obtain first near-end frequency domain information, and performing Fourier transform on the second near-end signal to obtain second near-end frequency domain information; performing filtering processing on the far-end frequency domain information by using the updated filter weight coefficient of the previous frame to obtain an echo frequency spectrum; subtracting the echo frequency spectrum from the first near-end frequency domain information to obtain a first initial error frequency spectrum; subtracting the echo frequency spectrum from the second near-end frequency domain information to obtain a second initial error frequency spectrum; performing centroid fitting on the first initial error frequency spectrum and the second initial error frequency spectrum respectively to obtain initial suppression gain factors of the first near-end signal and the second near-end signal, which comprises calculating a first cross-correlation coefficient between the far-end frequency domain information and the first initial error frequency spectrum, and a second cross-correlation coefficient between the far-end frequency domain information and the second initial error frequency spectrum, performing centroid fitting on the first cross-correlation coefficient to obtain a first initial suppression gain factor, and performing centroid fitting on the second cross-correlation coefficient to obtain a second initial suppression gain factor, multiplying the first initial suppression gain factor by the first initial error frequency spectrum to obtain the first

error frequency spectrum, and multiplying the second initial suppression gain factor by the second initial error frequency spectrum to obtain the second error frequency spectrum, and using the initial suppression gain factors as variable step-size factors for linear filtering processing of the next frame; performing residual echo suppression on the first error frequency spectrum and the second error frequency spectrum by using an adaptive zero-pole echo canceller to obtain a target frequency spectrum signal; after performing sub-band range selection on the initial suppression gain factors, obtaining a smooth factor, and performing full-band smoothing and exponential operation to obtain a secondary suppression gain factor; performing filtering processing on the target frequency spectrum signal by using the secondary suppression gain factor to obtain a target near-end voice signal.
